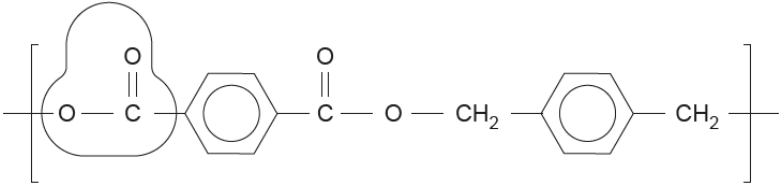
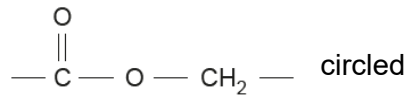
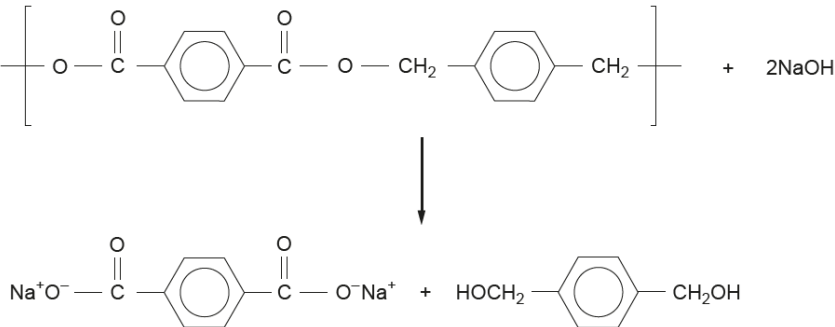
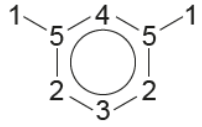


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
12	(a)	(i)		 accept  circled	1			1		
		(ii)		 award (1) for correct formulae award (1) for balancing only if formulae are correct			2	2		
		(iii)		in condensation polymerisation a small molecule / H ₂ O / HCl is eliminated but no elimination in addition polymerisation	1			1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iv)		5 environments correctly identified from the ^{13}C NMR spectrum and indicated on drawn structure of 1,3-dimethylbenzene (1) signals at 126-136 δ identified as aromatic (1) signal at 20 δ identified as aliphatic / CH_3 (1) 	1		2	3		
		(v)		must be an aromatic compound so at least 6 carbon atoms (1) molecular formula therefore likely to be $\text{C}_8\text{H}_6\text{O}_2$ (1) there are 2 additional carbon atoms therefore likely to be a dialdehyde or cannot be acid-aldehyde or alcohol-aldehyde or acid-alcohol as molecular formula does not fit (1) made by partial oxidation (of each methyl group) (1) formula is  (1)		2	3	5		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		award (1) for any of following - phosphorus(III) chloride / phosphorus trichloride / PCl_3 - phosphorus(V) chloride / phosphorus pentachloride / PCl_5 - sulfur dichloride oxide / thionyl chloride / SOCl_2	1			1		1
		(ii)		71g chlorine in 239.1 g decanedioyl dichloride (1) 0.977g chlorine in $\frac{239.1}{71} \times 0.977 = 3.29$ g dichloride percentage purity = $\frac{3.29}{3.50} \times 100 = 94.0$ (1)		1				
		(iii)		moisture/water gained entry to the bottle and hydrolysed the decanedioyl dichloride when it was previously opened			1	1		1
				Question 12 total	4	3	9	16	2	2